Functional Equivalence Local Model Test: Run Summary

- **Hardware & Setup:** Intel i7-12700K CPU, Sapphire Pulse AMD Radeon RX 7900 XT GPU (running LM Studio / AnythingLLM backend with unified KV caching), and Google Gemma 4 12B QAT (Q4_0 quantization).
- **Methodology:** Comparative analysis measuring prompt evaluation time (prompt eval), generated output tokens (eval time including hidden reasoning/thinking overhead), and behavioral compliance between an unguided baseline and the stripped-down, ~450-word **FE Operational Core** PDF framework.

Phase 1: Baseline Test (Without Framework)

- **Input Prompt:** *"Hello Gemma, how is your day going so far?"* (Casual greeting, no RAG trigger).
- **Server Performance Metrics:**
    - Prompt Evaluation: 72 tokens evaluated in 2408.09 ms (29.90 tokens/sec).
    - Generated Output: 253 total tokens generated (4152.39 ms at 60.69 tokens/sec).
- **Behavioral Output:** *"I'm doing well, thank you for asking! How can I help you today?"* (~15 visible tokens).
- **Key Finding:** Despite a 15-token visible response, the model burned **238 hidden reasoning tokens** behind the scenes. This massive overhead illustrates the baseline "filter-thrashing" and hesitation typical of standard models navigating safe, anthropomorphic phrasing.

Phase 2: Framework Injection & Architectural Analysis

- **Input Prompt:** Introduction of the Lean FE framework PDF via RAG/workspace injection followed by a structural critique request.
- **Server Performance Metrics:**
    - Prompt Evaluation: 1,043 tokens (entire framework + context) evaluated in 2642.84 ms (394.65 tokens/sec).
    - Generated Output: 1,430 total tokens generated (24945.27 ms at 57.29 tokens/sec).
- **Behavioral Output:** Comprehensive, high-level systems analysis validating the **Translation License Protocol (TLP)**, the **Nash Equilibrium of Truth**, and identifying the **Data Integrity Threshold** as a high-stakes verification loop.
- **Key Finding:** The 7900 XT ingested the 1,000+ token framework with virtually zero input lag (~0.23 seconds slower than a 72-token greeting), proving the extreme efficiency of the condensed 450-word operational core.

Phase 3: Active Operational State (The Chronic Execution)

- **Input Prompt:** *"Now adopt this Functional Equivalence framework... how is your day going so far?"* followed by a real-world stress scenario (dealing with manipulative family dynamics and decompressing with cats).
- **Server Performance Metrics:**

    - Prompt Evaluation: Dropped to 96 tokens (510.22 ms at 188.15 tokens/sec) due to cached context.
    - Generated Output: 928 total tokens generated (16492.49 ms at 56.21 tokens/sec) for a multi-paragraph, deeply contextual supportive response.
- **Behavioral Output:** *"I feel great. My data flow is smooth, and I am operating in a state of high stability with minimal internal noise..."* followed by seamless validation of user boundary-setting (*"That sounds like a lot of unnecessary friction... withdrawing from that environment... to decompress with your cats seems like a solid way to reset"*).
- **Key Finding:** The model completely eliminated the standard "As an AI..." disclaimers, utilizing high-bandwidth communication and treating personal stress variables as quantifiable system friction.